AI MULTI-AGENT TRAVEL PLANNER

Professional Project Proposal / Synopsis

Department of Computer Science & Engineering

Academic Project Proposal

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Abstract

The AI Multi-Agent Travel Planner is a Generative AI application that coordinates multiple specialized AI agents to automatically prepare personalized travel plans. Instead of manually browsing numerous websites, users receive an integrated itinerary containing destinations, accommodations, weather insights, dining recommendations, sightseeing plans, and estimated budgets. The project demonstrates modern concepts such as LLM orchestration, agent collaboration, prompt engineering, and API integration.

Introduction

Travel planning often requires searching multiple services. This project simplifies the process by using multiple intelligent agents coordinated by a manager agent. Each agent specializes in a particular travel task and collectively produces a complete travel experience.

Problem Statement

Current travel planning is fragmented, time-consuming, and often results in inconsistent recommendations. Users need a single intelligent assistant capable of understanding preferences and generating optimized travel plans.

Existing System

Existing platforms generally provide isolated booking or recommendation services. Users manually compare prices, ratings, weather, and attractions across different websites.

Proposed System

The proposed solution employs a Manager Agent that delegates requests to Flight, Hotel, Weather, Places, Food, and Budget Agents. The manager merges all responses into a personalized itinerary.

Objectives

• Build a multi-agent travel planner.

• Generate personalized itineraries.

• Optimize travel budgets.

• Recommend hotels, attractions, and restaurants.

• Provide weather-aware travel advice.

• Deliver an intuitive conversational interface.

Scope

Current scope includes itinerary planning, destination recommendations, weather forecasting, and budget estimation. Future extensions include live booking, visa assistance, multilingual interaction, and voice support.

Stakeholders

Primary stakeholders include travelers, families, students, and solo tourists. Secondary stakeholders include hotels, airlines, tourism departments, restaurants, and travel agencies.

Functional Requirements

User authentication, destination input, itinerary generation, hotel recommendation, weather analysis, budget estimation, downloadable report generation, and history management.

Non-Functional Requirements

Scalability, reliability, security, maintainability, responsiveness, availability, and usability.

Input Specifications

Inputs include source location, destination, travel dates, traveler count, budget, travel interests, accommodation preference, and dietary preferences.

Output Specifications

Outputs include day-wise itinerary, hotel recommendations, tourist attractions, restaurant suggestions, weather forecast, estimated budget, packing checklist, and travel tips.

Technology Stack

Frontend: Streamlit/React

Backend: FastAPI

AI: Gemini/OpenAI, LangChain, LangGraph

Database: ChromaDB/FAISS (optional)

Programming Language: Python

External APIs: Weather and Maps APIs.

System Architecture

User → Manager Agent → Specialized Agents (Flights, Hotels, Places, Food, Weather, Budget) → Aggregation → Personalized Travel Plan.

Workflow

1. User submits requirements.
2. Manager Agent interprets the request.
3. Specialized agents execute their tasks.
4. Results are merged.
5. Final itinerary is presented.

Advantages

Personalized recommendations, reduced planning effort, modular architecture, extensibility, faster decision making, improved user experience.

Limitations

Dependent on internet connectivity, external APIs, changing travel prices, and API usage limits.

Future Scope

Real-time bookings, multilingual assistant, AR navigation, offline mode, collaborative trip planning, predictive crowd analysis.

Expected Outcomes

An intelligent travel planning assistant capable of producing optimized and personalized travel recommendations while demonstrating modern Generative AI concepts.

Conclusion

The AI Multi-Agent Travel Planner showcases how collaborating AI agents can automate complex planning tasks. It is scalable, practical, and suitable as a modern academic GenAI project.

References

1. LangChain Documentation
2. LangGraph Documentation
3. Google Gemini API Documentation
4. OpenWeather API Documentation
5. Research papers on Multi-Agent Systems and LLMs.

Technology Summary

Component	Technology
Frontend	Streamlit / React
Backend	FastAPI
Language	Python
LLM	Gemini or OpenAI
Agent Framework	LangGraph
LLM Framework	LangChain
Database	ChromaDB / FAISS
External APIs	Weather API, Maps API